Marcelo Araújo Silva

Data Science | Fullstack Developer

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Experience

Full-Stack Developer — Aletheia (OBABIO 2026 System) 2026

- Implemented authentication and account security flows in Node.js/Express with Prisma ORM and PostgreSQL.
- Developed the REST API for user credentialing, including PDF report generation and transactional email templates (Handlebars) integrated via SMTP on the back-end.
- Collaborated on the front-end in React/TypeScript, and took part in the project's technical planning.

Trainee Backend Developer — TITAN 2026-Present

- Technologies: Node.js, NestJS, Prisma ORM, Git e GitHub
- Developing: Backend Applications, API, User Authentication

Data Scientist — Bioinformatics Lab (UFBA) 2022 – 2025

- Processed and performed quality control on **2,650+ samples** in Linux servers (**6+ TB**), generating standardized reports and outputs for downstream analysis.
- Developed modular tools in **Python** (scripts and CLIs) to automate multi-step workflows with clear input/output handling, focusing on maintainability and reproducibility.
- Conducted statistical analyses and visualizations using **R/Python** for scientific validation and communication (publication-ready figures and analytical narratives), supporting study conclusions and decision-making.
- Standardized environments and execution using **Conda** and **Docker**, ensuring portability across local machines and high-performance servers.
- Maintained version-controlled repositories (**Git/GitHub**) and technical documentation for collaboration and result traceability.

Software Engineering Projects

BioPoints — 1st Place, LIAO Hackathon @ UFBA (Full-Stack + ML) 2026

- Won **1st place** in a university hackathon themed around climate change, smart cities, and applied AI; conceived the product and led delivery end-to-end under strict time constraints.
- Architected and shipped the full **backend** (Node.js, NestJS, Prisma ORM, PostgreSQL): REST API, relational schema modeling multi-actor B2B workflows, and an ML-model serving endpoint.
- Built the **frontend** (React, Next.js) with an interactive interface that consumes the backend API in real time for on-the-fly computation and visualization.
- Authored the pitch deck and delivered the live presentation that secured the win against competing university teams.

Weather Data (Spring Boot, PostgreSQL, React) 2026 – Present

- Built a full-stack system that analyzes weather conditions on the user's birth date by integrating external geolocation and historical climate data APIs.
- Implemented a REST API with PostgreSQL persistence, consuming Open-Meteo and geocoding APIs while following repository/service layers, DTOs, and centralized exception handling.
- Developed climate analysis logic on the backend (averages, maximums, minimums, and textual classifications) and data visualization with dynamic charts on the React frontend using Recharts.

Technologies

Python, R, Bash; Java (Spring Boot); JavaScript/TypeScript (React, Next.js, Node.js, NestJS); Linux, Git/GitHub, Docker, Conda, Nextflow, PostgreSQL, Prisma ORM.

Open Source Contribution

2026 Open Source Contribution — Curation and analysis of a public dataset in Python (Brazilian companies and share capital), including version control and documentation on GitHub.

Certifications

2024 Duolingo English Test — Overall Score: 135
2023 Introduction to Computing — Institute of Exact Sciences, UFMG

Languages

Portuguese Native
English Advanced
Spanish Basic

Education

2026 – Present **B.S. in Computer Science** — UniJorge, Salvador-BA, Brazil
2021 – 2025 **B.S. in Biological Sciences** — Federal University of Bahia (UFBA), Salvador-BA, Brazil

Publications

2026 **Mycological Progress** — *Taxonomic and phylogenetic studies reveal three new species of the genus Geastrum in the Corollina section from Brazil.* Second author. DOI: 10.1007/s11557-025-02118-8